



LOKMANYA TILAK JANKALYAN SHIKSHAN SANSTHA'S
PRIYADARSHINI COLLEGE OF ENGINEERING



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

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Idea Submission Template / Code. Create. Innovate.

Team Name	Team Raksha
Problem Statement No.	2
Problem Statement Title	Improving Safety at Large Public Events
Team Leader Name	Shounak Joshi

THE PROBLEM

Guards watch screens. Danger builds unseen.

A control room watches dozens of flat camera feeds. A guard spots trouble only after the crowd stops moving. By then, crushing pressure is already passing body to body. Stillness is not the warning. It is the accident.

Kumbh 2025

2,751 cameras and AI alerts were live. The crush still killed 30.

Itaewon 2022

159 dead in a 3.2 metre alley. 11 calls ignored. No machine spoke up.

7 per m²

A crowd turns fluid. Shockwaves throw people metres. (Fruin)

Why stillness lies

A stopped crowd is not a warning. It is the accident. Pressure passes body to body long before feet stop. That is why cameras must read force, not faces.

Itaewon, 29 October 2022



18:34

First distress calls from the packed alley.



hours pass

Calls keep coming. No machine raises an alarm.



22:15

The alley locks. 159 dead, 196 injured.

Four hours of human warning. No machine ever paged anyone.

Danger is invisible pressure, not visible stillness.

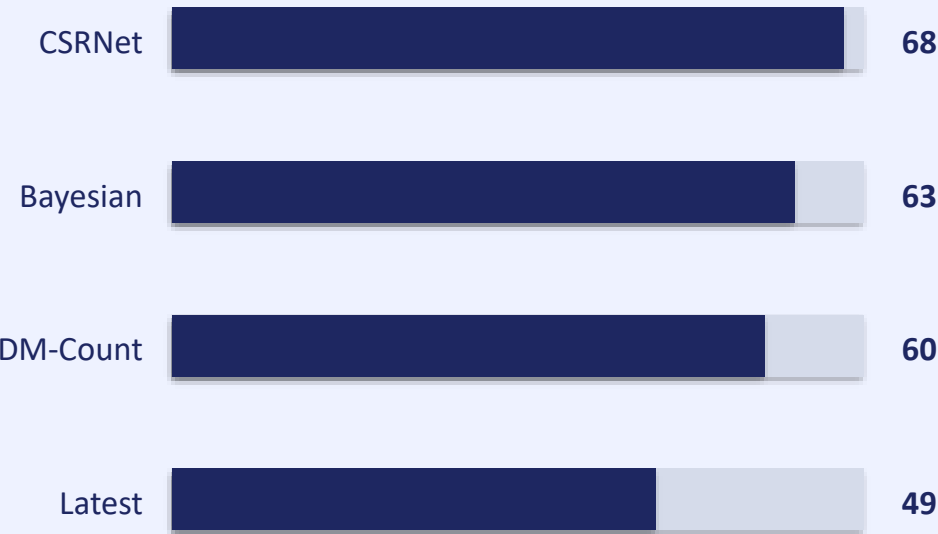
Measure pressure, warn early, open gates.

HOW IT WORKS



In plain words: map the ground right, count the crowd as heat, read its currents, warn on pressure.

Count error keeps falling (MAE, lower is better)



Heatmaps beat boxes

At crush density a detector scores F1 0.07 against 0.71 point-based. (NWPU-Crowd)

Pressure, not stillness

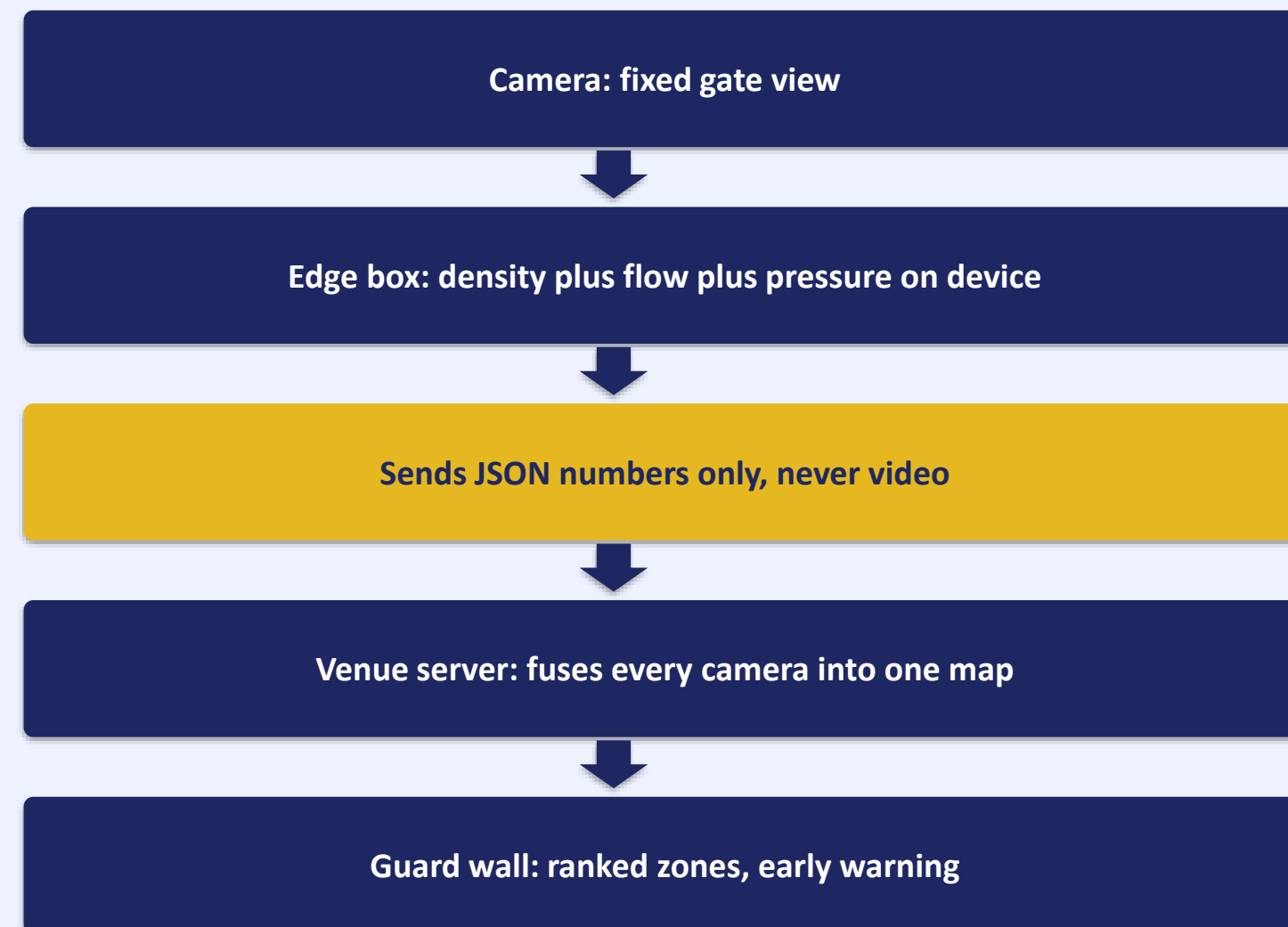
Pressure is density times uneven movement. Density or speed alone cannot warn. (Johansson 2008)

Side track, kept

Below 2 people per m2, box IDs still run and feed the dossier. Already built.

FEASIBILITY AND VIABILITY

One camera, one edge box



Raw video never leaves the camera box.

Video stays put

Each edge box sends small JSON numbers, never raw video. Footage is discarded at the camera.

Calibrated where it counts

Fixed gates are measured once. Other views use relative levels. No fake square metres.

Private by design

Heatmaps cannot show a face. Saved video is blurred, wiped after 7 days, zero identity in core.

Runs on site, not in the cloud

Edge box per 1 to 4 cameras (Jetson class)

Venue server fuses every feed into one map

Guard handsets get the alert, human confirms

Deployment scale numbers only, no laptop tests.

IMPACT AND BENEFITS

When pressure rises

Yellow watch

Guards watch the zone closely. Overflow gates get ready.

Red alert

Guards get steps on their handsets; a human confirms before any PA or door release.

Handover

A sealed evidence pack goes to the police.

Pilot targets: warn before critical, page guards in seconds, log every step.

Before

A wall of screens, one tired guard, and risks get missed.

After

Pressure rising at Gate B opens overflow at Gate A upstream. Security acts before the jam, not after.

Organizers

Spot a growing jam early, and tell guards which zone to clear.

Police

Every alert, zone and action exports as one evidence pack in a click.

Public

Faces stay blurred in saved video.

RESEARCH AND REFERENCES

The science behind it

Count the overlap

CSRNet (CVPR 2018), Bayesian Loss (ICCV 2019), DM-Count (NeurIPS 2020).
[arXiv:1802.10062](#) [arXiv:1908.03684](#) [DM-Count](#)

Map pixels to ground

Chan perspective maps (CVPR 2008), Zhang ground-plane fusion (CVPR 2019).
[people counting paper](#) [ground-plane paper](#)

Read the currents

Mehran social force (CVPR 2009), streaklines (ECCV 2010).
[CVPR 2009 pdf](#) [streakline project](#)

Warn on pressure

Helbing social force (1995), crowd disasters (2007), Johansson pressure (2008).
[DOI:10.1103/PhysRevE.51.4282](#) [arXiv:0708.3339](#) [pressure analysis](#)

Disasters, limits, footage

Itaewon and Love Parade

159 dead, Seoul 2022, ignored calls. 21 dead, Duisburg 2010. Turbulence, not panic.

Kumbh and Hathras

30 dead, Jan 2025, with AI live. 121 dead, Jul 2024, 80k permit vs 250k crowd.

Danger thresholds

Fluid at 7 per m2. Serious above 4. Flow peaks at 2 to 3. (Fruin, Still)

Test footage

Public clips: station, crossing, temple.
[Wikimedia Commons](#)